

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

%matplotlib inline

data = pd.read_csv("pca_variable_loadings.csv")

print(data.shape)
data.head()
```

```
(9, 4)
```

	Variable	PC1	PC2	PC3
0	Climate	0.190	0.017	0.207
1	Housing	0.544	0.020	0.204
2	Health	0.782	-0.605	0.144
3	Crime	0.365	0.294	0.585
4	Transportation	0.585	0.085	0.234

```
df = data[["PC1", "PC2", "PC3"]]
```

```
print(df)
```

	PC1	PC2	PC3
0	0.190	0.017	0.207
1	0.544	0.020	0.204
2	0.782	-0.605	0.144
3	0.365	0.294	0.585
4	0.585	0.085	0.234
5	0.394	-0.273	0.027
6	0.985	0.126	-0.111
7	0.520	0.402	0.519
8	0.142	0.150	0.239

```
for column in df.columns:
    print("\n", column)
    print(data.loc[df[column].abs().idxmax(), "Variable"])
    print("Loading:", df[column].abs().max())
```

```
PC1
Arts
Loading: 0.985
```

```
PC2
Health
Loading: 0.605
```

```
PC3
Crime
Loading: 0.585

loading_values = df.abs()

max_value = loading_values.max().max()

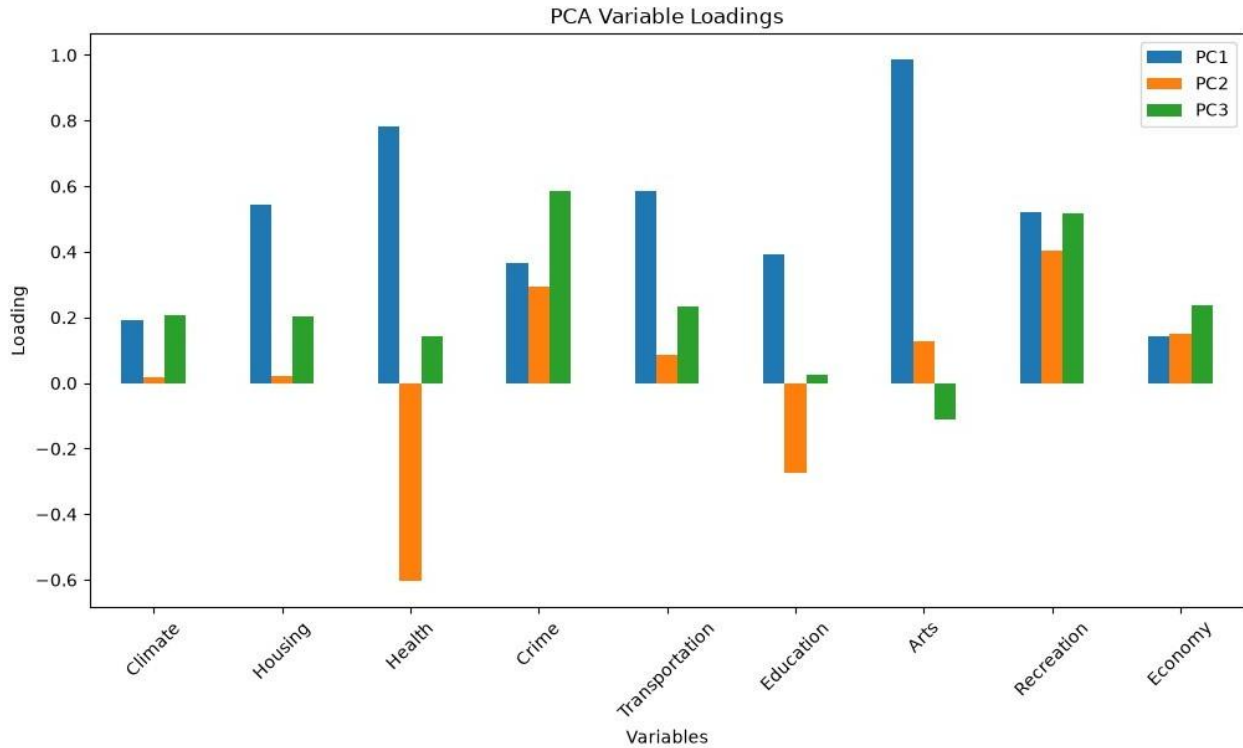
row, column = loading_values.stack().idxmax()

print("Variable:", data.loc[row, "Variable"])
print("Principal Component:", column)
print("Loading:", df.loc[row, column])

Variable: Arts
Principal Component: PC1
Loading: 0.985

data.set_index("Variable")[["PC1", "PC2", "PC3"]].plot(
    kind="bar",
    figsize=(12, 6)
)

plt.title("PCA Variable Loadings")
plt.xlabel("Variables")
plt.ylabel("Loading")
plt.xticks(rotation=45)
plt.show()
```



INFERENCE:

The PCA loading values were analyzed to identify the variable with the largest contribution to each principal component.

Arts has the highest loading in PC1 with a value of 0.985.

Health has the highest absolute loading in PC2 with a value of -0.605.

Crime has the highest loading in PC3 with a value of 0.585.

Therefore, Arts is the variable that contributes most strongly to the principal components overall because it has the largest absolute loading of 0.985.